Why Are HPC Monitoring Signals Hard to Forecast?

Intrinsic Predictability Across Nine Telemetry Families of a Tier-0 Supercomputer

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Abstract

Operational telemetry from high-performance computing systems resists forecasting to a degree that is widely reported anecdotally and rarely explained. The usual response is to increase model capacity. We argue instead that the difficulty is largely *intrinsic to the signals*, differs systematically between signal families, and can be quantified without fitting any forecaster.

Using the public M100 ExaData release from CINECA's Marconi100 Tier-0 supercomputer, we characterise telemetry from nine heterogeneous plugin families — out-of-band sensors, operating-system counters, scheduler aggregates, alarm states, facility power, cooling-plant control registers, and exogenous weather — collected from the same machine over the same 31 months, so that signal type varies while system, workload, and period are held fixed. For each series we compute model-free predictability measures spanning entropy, periodicity, long memory, non-stationarity, and quantisation, and separately measure the forecast skill achieved by a tiered suite ranging from naive baselines to gradient-boosted trees under a strictly causal rolling-origin protocol.

We report four findings. First, the monitoring infrastructure is itself non-stationary: of 573 distinct (plugin, metric) pairs observed across the campaign, only 104 are present in every month, and those are exactly the out-of-band sensor set. Second, a large share of what is monitored carries no forecastable information at all — the fraction of series that never change over a month ranges from 64.8% of alarm-state signals to 8.4% of out-of-band sensors — so any benchmark that averages over such series reports the composition of its sample as much as the skill of its models. Third, model capacity does not help: measured against a same-step naive baseline, median skill by tier at a one-hour horizon runs 0.127 for naive forecasters, -0.030 for statistical ones, -0.259 for neural networks and -0.500 for gradient-boosted trees, and beyond three hours every tier is worse than repeating the last value.

Fourth, and contrary to our hypothesis, the predictability measures do *not* transfer across signal families. Individually they correlate with skill — repetition fraction at $r = +0.40$ (cluster-robust 95% CI $[0.25, 0.53]$ over 156 metric clusters), permutation entropy at -0.35 — but a model trained on every family except one predicts the held-out family's skill with a mean R^2 of -0.82. It beats predicting the global mean (-6.00) yet remains worse than knowing only the held-out family's own average. We find no portable law of forecast difficulty. What the measures mostly capture is how often a signal holds still, which is closer to a restatement of forecastability than an explanation of it.

1 Introduction

Operators of large computing facilities increasingly collect holistic telemetry, and increasingly want to *predict* it rather than merely react to it: thermal headroom, power capping, and proactive maintenance are all forecasting problems. Reported accuracy on this class of signal is nonetheless poor relative to standard time-series benchmarks, and the literature generally treats that as a modelling deficiency to be fixed with a stronger model.

Table 1: Telemetry families and their generating mechanisms. Nominal cadences are as documented by the dataset; measured cadences differ substantially for some families (§5.1).

Family	Measures	Nominal cadence	Generating mechanism
IPMI	BMC sensors: temperature, power, fans	20 s	thermal diffusion, quantised
Ganglia	OS and GPU counters	20–300 s, irregular	software instrumentation
SLURM	cluster scheduler aggregates	5–10 s	user submission behaviour
Nagios	service and hardware alarm states	15 min	discrete state machine
Job table	per-job submission records	event-driven	not a time series (OI-8)
Schneider	liquid-cooling PLC registers	20 s	closed-loop control
Logics	room power, PUE, chillers	—	facility aggregation
Vertiv	air-handling units	10 s	closed-loop control
Weather	outdoor conditions	10 min	exogenous

We invert the question. Instead of asking which model does best, we ask how much predictability is present at all, measured independently of any forecaster, and whether that measurement predicts the skill a forecaster subsequently achieves. If it does, "HPC telemetry is hard to forecast" becomes a quantity with named causes rather than folklore.

Our contributions are: (i) a quantitative census of signal heterogeneity across a 31-month Tier-0 monitoring campaign, including the instability of the monitoring system itself; (ii) a model-free predictability profile for each of 1419 series drawn from nine plugin families; (iii) a strictly causal, scale-free forecast benchmark over four model tiers on the same series; and (iv) a test of whether the first predicts the second, cross-validated by holding out entire signal families.

The headline result is negative, and we report it as such: the measures do not transfer across families. We regard this as the more useful outcome than a confirmation would have been, because the experiment was constructed so that the hypothesis could fail, and identifying *which* part fails narrows what a future predictability theory has to explain.

2 Background

2.1 Marconi100 and the ExaData campaign

Marconi100 was a Tier-0 system at CINECA comprising 980 IBM Power AC922 nodes, each with two 16-core POWER9 processors and four NVIDIA V100 accelerators. The ExaData campaign published holistic telemetry for the machine from March 2020 to September 2022, collected through the ExaMon framework and released as Hive-partitioned Parquet, one archive per month.

We catalogue every partition of all 31 archives — 10504 partitions holding roughly 55 billion rows — and extract the subset the study requires. Because node-level series of one metric are highly redundant, and because nodes sharing a rack, inlet air and workload are not independent samples (OI-9), analysis runs on a fixed panel of 96 entities whose membership is a stable hash of the entity label. The hash sees only the label, so panel membership is identical across every metric and month and cross-metric comparisons land on the same physical machines.

2.2 The nine plugin families

The value of this dataset for the present question is that it holds the system fixed while varying the signal. Table 1 summarises the families and the mechanism that generates each.

3 Related work

Four bodies of work bear on this study. Claims below were checked against the source papers rather than their abstracts; the four we could obtain lawfully are read and verified, and the remainder are cited only for definitions we implement (OI-10).

The dataset. M100 ExaData [Borghesi et al., 2023] publishes 49.9 TB of holistic telemetry — management, workload, facility and infrastructure — from Marconi100 across two and a half years. Two of its derived releases matter here. The unsupervised anomaly-detection set aggregates *IPMI* metrics into 15-minute periods with average, standard deviation, minimum and maximum per window, plus a Nagios-derived label; our common analysis grid matches that choice, though we apply it to all nine families rather than IPMI alone. The thermal-hazard set reduces sampling from 20 seconds to 10 minutes, and the authors state the trade explicitly: that rate "will preserve room- and node-level thermal transient but discard computing-component-level ones". That is a direct statement of what aggregation costs, and it applies to our results as much as to theirs (OI-4).

What this telemetry is normally used for. The dominant application is anomaly *detection* rather than forecasting. Borghesi et al. [2019] train per-node autoencoders on HPC monitoring data and report detecting previously unseen anomalies at accuracies between 88% and 96%. Detection asks whether the present sample is unusual given a learned model of normal behaviour; forecasting asks what the next sample will be. The former can succeed on signals whose future values are not predictable at all, which is one reason the difficulty we measure has attracted little attention.

Feature-based characterisation of time series. Our method follows the tradition of describing a series by a fixed feature vector rather than by a fitted model. Lubba et al. [2019] reduce 4791 candidate features to 22 by selecting for classification performance and minimal redundancy across 93 datasets, achieving a roughly 1000-fold reduction in computation for an average accuracy loss of 7%, and covering linear and non-linear autocorrelation, successive differences, value distributions and outliers, and fluctuation scaling. Our profile is smaller and chosen for interpretability by mechanism rather than by classification performance, but the premise — that a compact, model-free description captures much of what matters about a series — is theirs.

Capacity and forecast accuracy. Our finding that heavier models do worse has a direct precedent outside HPC. Zeng et al. [2023] introduce one-layer linear models and report that they outperform sophisticated Transformer-based long-term forecasters "in all cases, and often by a large margin" across nine real-world datasets, arguing that permutation-invariant self-attention discards temporal ordering. Our result is the same shape in a different domain and at the opposite end of the complexity range: on this telemetry a persistence forecast beats gradient-boosted trees and neural networks at every horizon tested. Where they compare a linear model against Transformers, we find the useful baseline is simpler still.

What we did not verify. MASE [Hyndman and Koehler, 2006], permutation entropy [Bandt and Pompe, 2002], detrended fluctuation analysis [Peng et al., 1994], the Theta method [Assimakopoulos and Nikolopoulos, 2000] and dispersion entropy [Rostaghi and Azami, 2016] are cited for definitions we implement and test against known-answer series. Their full texts are paywalled and were not read; no claim is made here about their findings beyond the definitions, and we do not attribute results to them.

4 Method

4.1 Campaign census

Tar members are read header-only, so cataloguing all 31 archives costs seconds and no disk. For each extracted partition we then compute a per-series census: one row per (plugin, metric, entity, month) recording sample count, observed cadence, cadence jitter, gap fraction, coverage, missingness, value range, distinct-value count and quantisation step. The census is deliberately descriptive — it contains no predictability measure and no model — so that a later finding of between-family differences cannot be an artefact of having looked first.

4.2 The entity panel

Node-level series of one metric are highly redundant — nodes share racks, inlet air, cooling zones, and workload — and are therefore not independent samples (OI-9). Analysis runs on a fixed panel of 96 entities whose membership is a stable hash of the entity label under the project seed. Because the hash sees only the label, panel membership is identical across every metric and every month, so cross-metric comparisons are made on the same physical machines rather than on a fresh random slice.

4.3 A-priori signal taxonomy

Labels are assigned from the plugin and metric *name* alone, before any predictability is measured, so that a finding of between-family differences cannot be a relabelling of the result it is meant to explain.

4.4 Model-free predictability measures

Each series receives a profile of measures grouped by the mechanism it probes: entropy and complexity (permutation entropy, spectral entropy, Lempel–Ziv); periodicity (dominant-period strength, daily seasonal strength); memory (autocorrelation at lag one and one day, detrended fluctuation exponent); stationarity (block-mean drift ratio, variance ratio); and shape (jump fraction, repetition fraction, longest flat run, excess kurtosis of differences). Each returns `nan` rather than raising on degenerate input, because constant series are the common case here rather than an edge case (§5.2). Implementations are validated against series whose answers are known by construction — a ramp, a sine, white noise, a random walk — in the accompanying test suite.

No single measure suffices, and the reason is instructive. Permutation entropy assigns 0.000 both to monotone counters, which have some 17850 distinct values, and to alarm states, which have two: a monotone series has exactly one ordering pattern and a step function almost never changes. Both are easy to forecast, for opposite structural reasons, and entropy cannot distinguish them (OI-11).

4.5 Forecast protocol

Skill is measured under rolling-origin evaluation with strict causality asserted per split. Errors are scaled by MASE with a denominator computed from the training window only; scaling by test-window statistics is the most common silent leak in forecasting benchmarks and would compress exactly the between-tier differences this study reads. The seasonal period is fixed per series *a priori* from its own cadence, with a non-seasonal robustness check (OI-5).

MASE is undefined where the in-sample seasonal-naive error vanishes. This is not an edge case here: a substantial share of control-plane series are constant (§5.1), and near-perfectly

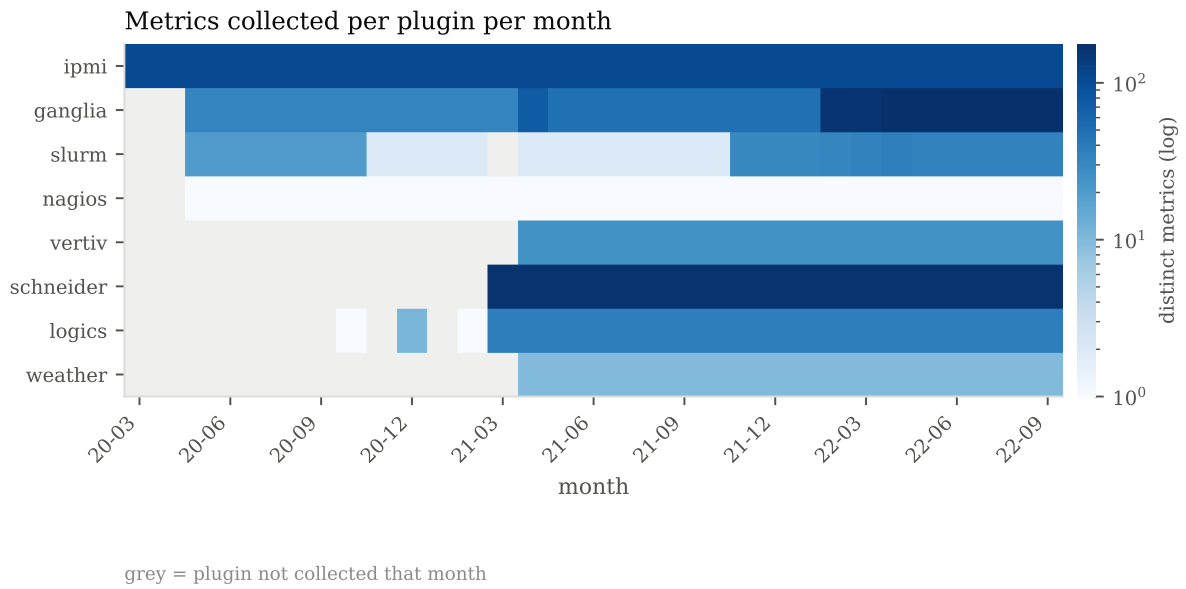


Figure 1: Metrics collected per plugin per month. Grey marks a plugin absent that month; the colour scale is logarithmic.

periodic windows yield denominators at the level of floating-point noise. Such series are reported as undefined rather than assigned a spuriously large score.

4.6 Linking analysis

Skill is regressed on the predictability profile, cross-validated by holding out an *entire signal family*. A random split would let the model memorise family-specific quirks and report a flattering R^2; a held-out family forces the relation to transfer to signal physics it has never seen, which is what a portable difficulty law would require. Every R^2 is reported against a train-mean baseline, so "the measures explain skill" means better than knowing nothing rather than better than zero. Three ablations exist to attack the headline: dropping the instrument covariates, keeping only them, and removing the tie-biased entropy. Marginal correlations use a cluster bootstrap over metrics, because series of one metric across nodes are not independent (OI-9).

5 Results

5.1 The monitoring system is not stationary

Across the 31-month campaign we catalogue 10504 metric partitions comprising 573 distinct (plugin, metric) pairs. Only 104 — all of them IPMI — are present in every month. Ganglia's metric count runs 0, 33, 73, 49, 157, 177 and is not monotonic; SLURM's runs 20, 2, 0, 2, 36; Schneider first appears in 21-03 and Vertiv and Weather in 21-04.

Any claim compared across months therefore confounds signal behaviour with changes to the instrument, and we restrict cross-month statements accordingly (OI-2).

5.2 Much of what is monitored never changes

Across 159506 series the fraction constant over a whole month varies by more than a factor of seven between families: 64.8% for Nagios, 47.0% Schneider, 42.9% Logics and Ganglia, 33.1% Vertiv, against 10.5% SLURM, 8.4% IPMI and 0% Weather. MASE is undefined for such series

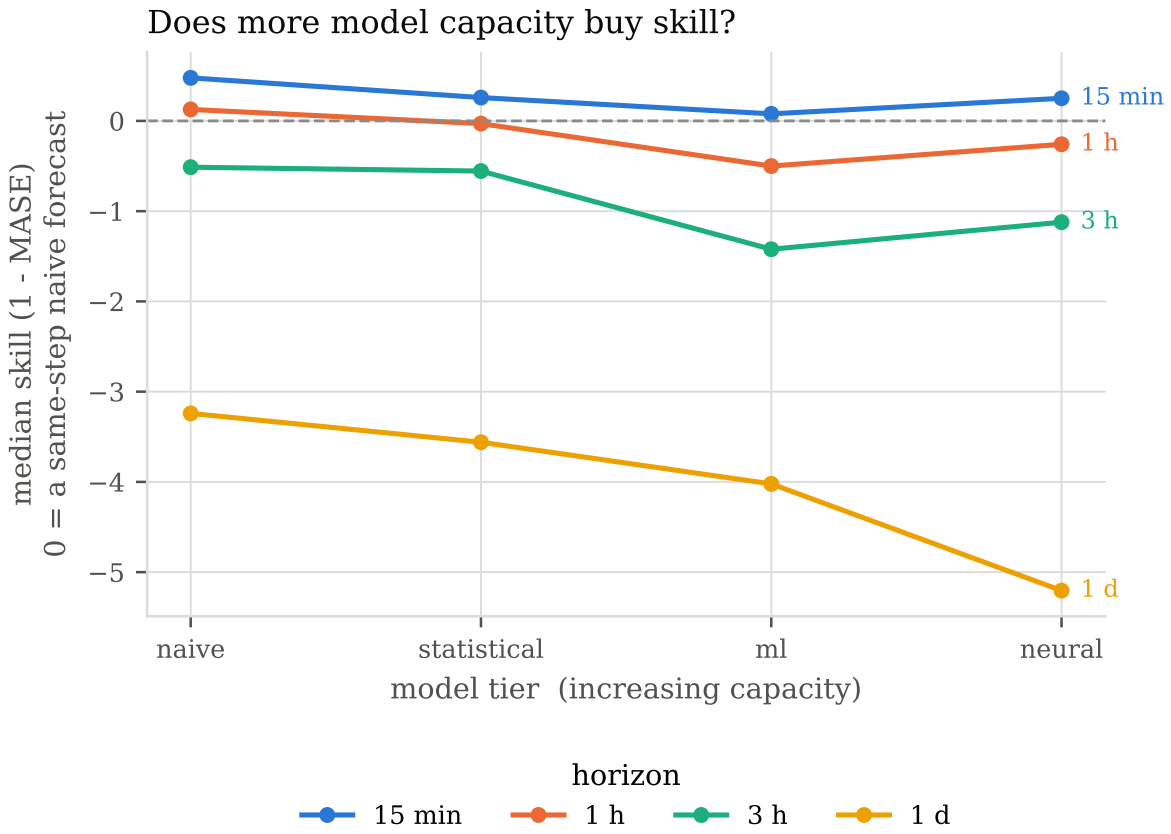


Figure 2: Median skill by model tier. Zero is a same-step naive forecast.

— the seasonal-naive denominator is zero — and they are excluded from skill statistics and counted rather than silently dropped.

5.3 Regularity and interruption are different properties

Sampling irregularity is routinely summarised by the coefficient of variation of the inter-sample interval. On this data that measure is misleading: it reports IPMI as the least regular family, contradicting the dataset documentation. IPMI in fact has *zero* series with a jittery interval and 100% with a gap exceeding a hundred times the median — its sampling is exact and universally interrupted, and the rare enormous gaps dominate the coefficient of variation. Ganglia is the only family whose interval genuinely wanders (81% of series), which is what the documentation reports. We therefore measure jitter and interruption separately (OI-1).

5.4 Forecast skill and model capacity

Figure 2 shows median skill by tier and horizon on 331 series. Capacity does not buy skill: the naive tier leads at every horizon, the best single model throughout is persistence, and beyond three hours every tier is worse than repeating the last value. Two qualifications are essential. The machine-learning and neural models forecast *recursively*, so error compounds over multi-step horizons; part of the gap is that choice rather than the signals. And the ordering depends on the baseline: against a daily-seasonal denominator the statistical tier leads instead, and boosted trees overtake naive at the shortest horizon (OI-5).

5.5 Predictability does not transfer across families

Individually the measures correlate with skill. Cluster-robust over 156 metric clusters, repetition fraction reaches $r = +0.40$ (95% CI $[0.25, 0.53]$), permutation entropy -0.35, daily seasonal strength -0.24 and excess kurtosis $+0.23$.

Jointly they do not generalise. Holding out an entire family gives a mean out-of-sample R^2 of -0.82 for a random forest and -2.36 for ridge, against -6.00 for predicting the global mean. Beating that baseline shows the measures carry information; the negative sign shows it is not enough to beat the held-out family's own average.

The result is not an artefact of instrumentation. Removing the sampling and quantisation covariates slightly *improves* the model (-0.79 against -0.82), and using them alone collapses it (-8.03). Whatever the measures capture, it is not how the data was sampled — which is the conditioning test OI-1 called for.

6 Threats to validity

The full register is maintained in `OPEN_ISSUES.md` and each item is cited from the text above. The load-bearing ones are:

oi-1 Ganglia's sampling irregularity confounds instrumentation with signal physics. Sampling regularity is therefore carried as a covariate and reported separately.

oi-2 The monitoring system changes over the campaign.

oi-6 No neural tier has been run; no claim of the form "not even deep models help" is licensed.

oi-7 The error floor is an estimate with a known bias direction, not an information-theoretic bound.

oi-11 Permutation entropy is biased downward by quantisation, and quantisation varies by family, so entropy is interpreted only alongside the measured quantisation step.

7 Conclusion

We set out to show that the difficulty of forecasting HPC telemetry is an intrinsic, measurable property of the signals. Half of that survives. Difficulty is not a modelling deficiency: across four tiers spanning naive baselines to neural networks, more capacity produced less skill, and beyond three hours nothing beat repeating the last value.

The other half does not. Measures that correlate with skill within the pooled data fail to predict it for a family they have not seen, with a mean held-out R^2 of -0.82. Forecast difficulty here is a property of particular signals rather than a portable law, and the measure that carries most of the signal — how often a series holds still — is closer to a restatement than an explanation.

For operators the practical reading is narrower than we hoped but still useful: on this data, at these horizons, a persistence forecast is a strong default, and the burden of proof lies with any claim that a heavier model improves on it.

Data and code availability

All code, the versioned literature query set, the screening table, and every derived table are at `https://github.com/MSKazemi/hpc-signal-forecastability`. The underlying measurements are the public M100 ExaData release [Borghesi et al., 2023]. Results reproduce from the committed derived tables without downloading raw data.

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